After reviewing this checklist, what will you do to become confident for all objectives?

4.3 Multiply and Divide Mixed Numbers and Complex Fractions

Learning Objectives

By the end of this section, you will be able to:

- Multiply and divide mixed numbers
- Translate phrases to expressions with fractions
- Simplify complex fractions
- Simplify expressions written with a fraction bar

BE PREPARED 4.6 Before you get started, take this readiness quiz.

Divide and reduce, if possible: $(4+5)\div(10-7)$.
If you missed this problem, review Example 3.21.

BE PREPARED 4.7 Multiply and write the answer in simplified form: $\frac{1}{8}\cdot\frac{2}{3}$.
If you missed this problem, review Example 4.25.

BE PREPARED 4.8 Convert $2\frac{3}{5}$ into an improper fraction.
If you missed this problem, review Example 4.11.

Multiply and Divide Mixed Numbers

In the previous section, you learned how to multiply and divide fractions. All of the examples there used either proper or improper fractions. What happens when you are asked to multiply or divide mixed numbers? Remember that we can convert a mixed number to an improper fraction. And you learned how to do that in Visualize Fractions.

EXAMPLE 4.37

Multiply: $3\frac{1}{3}\cdot\frac{5}{8}$

Solution

	$3\frac{1}{3}\cdot\frac{5}{8}$
Convert $3\frac{1}{3}$ to an improper fraction.	$\frac{10}{3}\cdot\frac{5}{8}$
Multiply.	$\frac{10\cdot5}{3\cdot8}$
Look for common factors.	$\frac{2\cdot5\cdot5}{3\cdot2\cdot4}$
Remove common factors.	$\frac{5\cdot5}{3\cdot4}$
Simplify.	$\frac{25}{12}$

Notice that we left the answer as an improper fraction, $\frac{25}{12}$, and did not convert it to a mixed number. In algebra, it is preferable to write answers as improper fractions instead of mixed numbers. This avoids any possible confusion between $2\frac{1}{12}$ and $2\cdot\frac{1}{12}$.

TRY IT 4.73 Multiply, and write your answer in simplified form: $5\frac{2}{3}\cdot\frac{6}{17}$.

TRY IT 4.74 Multiply, and write your answer in simplified form: $\frac{3}{7} \cdot 5\frac{1}{4}$.



HOW TO

Multiply or divide mixed numbers.

Step 1. Convert the mixed numbers to improper fractions.
Step 2. Follow the rules for fraction multiplication or division.
Step 3. Simplify if possible.

EXAMPLE 4.38

Multiply, and write your answer in simplified form: $2\frac{4}{5}\left(-1\frac{7}{8}\right)$.

Solution

	$2\frac{4}{5}\left(-1\frac{7}{8}\right)$
Convert mixed numbers to improper fractions.	$\frac{14}{5}\left(-\frac{15}{8}\right)$
Multiply.	$-\frac{14 \cdot 15}{5 \cdot 8}$
Look for common factors.	$-\frac{2 \cdot 7 \cdot 5 \cdot 3}{5 \cdot 2 \cdot 4}$
Remove common factors.	$-\frac{7 \cdot 3}{4}$
Simplify.	$-\frac{21}{4}$

TRY IT 4.75 Multiply, and write your answer in simplified form. $5\frac{5}{7}\left(-2\frac{5}{8}\right)$.

TRY IT 4.76 Multiply, and write your answer in simplified form. $-3\frac{2}{5} \cdot 4\frac{1}{6}$.

EXAMPLE 4.39

Divide, and write your answer in simplified form: $3\frac{4}{7} \div 5$.

Solution

	$3\frac{4}{7} \div 5$
Convert mixed numbers to improper fractions.	$\frac{25}{7} \div \frac{5}{1}$
Multiply the first fraction by the reciprocal of the second.	$\frac{25}{7} \cdot \frac{1}{5}$
Multiply.	$\frac{25 \cdot 1}{7 \cdot 5}$

Look for common factors.	$\frac{\cancel{7}\cdot 5\cdot 1}{7\cdot\cancel{7}}$
Remove common factors.	$\frac{5\cdot 1}{7}$
Simplify.	$\frac{5}{7}$

TRY IT 4.77 Divide, and write your answer in simplified form: $4\frac{3}{8} \div 7$.

TRY IT 4.78 Divide, and write your answer in simplified form: $2\frac{5}{8} \div 3$.

EXAMPLE 4.40

Divide: $2\frac{1}{2} \div 1\frac{1}{4}$.

Solution

	$2\frac{1}{2} \div 1\frac{1}{4}$
Convert mixed numbers to improper fractions.	$\frac{5}{2} \div \frac{5}{4}$
Multiply the first fraction by the reciprocal of the second.	$\frac{5}{2} \cdot \frac{4}{5}$
Multiply.	$\frac{5\cdot 4}{2\cdot 5}$
Look for common factors.	$\frac{\cancel{5}\cdot\cancel{2}\cdot 2}{\cancel{2}\cdot 1\cdot\cancel{5}}$
Remove common factors.	$\frac{2}{1}$
Simplify.	2

TRY IT 4.79 Divide, and write your answer in simplified form: $2\frac{2}{3} \div 1\frac{1}{3}$.

TRY IT 4.80 Divide, and write your answer in simplified form: $3\frac{3}{4} \div 1\frac{1}{2}$.

Translate Phrases to Expressions with Fractions

The words *quotient* and *ratio* are often used to describe fractions. In Subtract Whole Numbers, we defined quotient as the result of division. The quotient of a and b is the result you get from dividing a by b, or $\frac{a}{b}$. Let's practice translating some phrases into algebraic expressions using these terms.

EXAMPLE 4.41

Translate the phrase into an algebraic expression: "the quotient of $3x$ and 8 ."

Solution

The keyword is *quotient*; it tells us that the operation is division. Look for the words *of* and *and* to find the numbers to divide.

$$\text{The quotient } \textbf{of } 3x \textbf{ and } 8.$$

This tells us that we need to divide $3x$ by 8. $\frac{3x}{8}$

TRY IT 4.81 Translate the phrase into an algebraic expression: the quotient of $9s$ and 14.

TRY IT 4.82 Translate the phrase into an algebraic expression: the quotient of $5y$ and 6.

EXAMPLE 4.42

Translate the phrase into an algebraic expression: the quotient of the difference of m and n, and p.

Solution

We are looking for the *quotient* of the *difference* of m and n, and p. This means we want to divide the difference of m and n by p.

$$\frac{m-n}{p}$$

TRY IT 4.83 Translate the phrase into an algebraic expression: the quotient of the difference of a and b, and cd.

TRY IT 4.84 Translate the phrase into an algebraic expression: the quotient of the sum of p and q, and r.

Simplify Complex Fractions

Our work with fractions so far has included proper fractions, improper fractions, and mixed numbers. Another kind of fraction is called **complex fraction**, which is a fraction in which the numerator or the denominator contains a fraction.

Some examples of complex fractions are:

$$\frac{\frac{6}{7}}{3} \qquad \frac{\frac{3}{4}}{\frac{5}{8}} \qquad \frac{\frac{x}{2}}{\frac{5}{6}}$$

To simplify a complex fraction, remember that the fraction bar means division. So the complex fraction $\frac{\frac{3}{4}}{\frac{5}{8}}$ can be written as $\frac{3}{4} \div \frac{5}{8}$.

EXAMPLE 4.43

Simplify: $\frac{\frac{3}{4}}{\frac{5}{8}}$.

Solution

	$\frac{\frac{3}{4}}{\frac{5}{8}}$
Rewrite as division.	$\frac{3}{4} \div \frac{5}{8}$
Multiply the first fraction by the reciprocal of the second.	$\frac{3}{4} \cdot \frac{8}{5}$
Multiply.	$\frac{3 \cdot 8}{4 \cdot 5}$
Look for common factors.	$\frac{3 \cdot \cancel{4} \cdot 2}{\cancel{4} \cdot 5}$
Remove common factors and simplify.	$\frac{6}{5}$

TRY IT 4.85 Simplify: $\frac{\frac{2}{3}}{\frac{5}{6}}$.

TRY IT 4.86 Simplify: $\frac{\frac{3}{7}}{\frac{6}{11}}$.

HOW TO

Simplify a complex fraction.

Step 1. Rewrite the complex fraction as a division problem.
Step 2. Follow the rules for dividing fractions.
Step 3. Simplify if possible.

EXAMPLE 4.44

Simplify: $\frac{-\frac{6}{7}}{3}$.

Solution

	$\frac{-\frac{6}{7}}{3}$
Rewrite as division.	$-\frac{6}{7} \div 3$
Multiply the first fraction by the reciprocal of the second.	$-\frac{6}{7} \cdot \frac{1}{3}$
Multiply; the product will be negative.	$-\frac{6 \cdot 1}{7 \cdot 3}$

Look for common factors.	$-\frac{3\cdot 2\cdot 1}{7\cdot 3}$
Remove common factors and simplify.	$-\frac{2}{7}$

TRY IT 4.87 Simplify: $\frac{-\frac{8}{7}}{4}$.

TRY IT 4.88 Simplify: $-\frac{3}{\frac{9}{10}}$.

EXAMPLE 4.45

Simplify: $\frac{\frac{x}{2}}{\frac{xy}{6}}$.

Solution

	$\frac{\frac{x}{2}}{\frac{xy}{6}}$
Rewrite as division.	$\frac{x}{2} \div \frac{xy}{6}$
Multiply the first fraction by the reciprocal of the second.	$\frac{x}{2} \cdot \frac{6}{xy}$
Multiply.	$\frac{x\cdot 6}{2\cdot xy}$
Look for common factors.	$\frac{x\cdot 3\cdot 2}{2\cdot x\cdot y}$
Remove common factors and simplify.	$\frac{3}{y}$

TRY IT 4.89 Simplify: $\frac{\frac{a}{8}}{\frac{ab}{6}}$.

TRY IT 4.90 Simplify: $\frac{\frac{p}{2}}{\frac{pq}{8}}$.

EXAMPLE 4.46

Simplify: $\frac{2\frac{3}{4}}{\frac{1}{8}}$.

Solution

	$\frac{2\frac{3}{4}}{\frac{1}{8}}$
Rewrite as division.	$2\frac{3}{4} \div \frac{1}{8}$
Change the mixed number to an improper fraction.	$\frac{11}{4} \div \frac{1}{8}$
Multiply the first fraction by the reciprocal of the second.	$\frac{11}{4} \cdot \frac{8}{1}$
Multiply.	$\frac{11 \cdot 8}{4 \cdot 1}$
Look for common factors.	$\frac{11 \cdot \cancel{4} \cdot 2}{\cancel{4} \cdot 1}$
Remove common factors and simplify.	22

TRY IT 4.91 Simplify: $\frac{\frac{5}{7}}{1\frac{2}{5}}$.

TRY IT 4.92 Simplify: $\frac{\frac{8}{5}}{3\frac{1}{5}}$.

Simplify Expressions with a Fraction Bar

Where does the negative sign go in a fraction? Usually, the negative sign is placed in front of the fraction, but you will sometimes see a fraction with a negative numerator or denominator. Remember that fractions represent division. The fraction $-\frac{1}{3}$ could be the result of dividing $\frac{-1}{3}$, a negative by a positive, or of dividing $\frac{1}{-3}$, a positive by a negative. When the numerator and denominator have different signs, the quotient is negative.

$\frac{-1}{3} = -\frac{1}{3}$ $\frac{\text{negative}}{\text{positive}} = \text{negative}$ $\frac{1}{-3} = -\frac{1}{3}$ $\frac{\text{positive}}{\text{negative}} = \text{negative}$

If *both* the numerator and denominator are negative, then the fraction itself is positive because we are dividing a negative by a negative.

$$\frac{-1}{-3} = \frac{1}{3} \qquad \frac{\text{negative}}{\text{negative}} = \text{positive}$$

Placement of Negative Sign in a Fraction

For any positive numbers a and b,

$$\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$$

EXAMPLE 4.47

Which of the following fractions are equivalent to $\frac{7}{-8}$?

$$\frac{-7}{-8}, \frac{-7}{8}, \frac{7}{8}, -\frac{7}{8}$$

Solution

The quotient of a positive and a negative is a negative, so $\frac{7}{-8}$ is negative. Of the fractions listed, $\frac{-7}{8}$ and $-\frac{7}{8}$ are also negative.

TRY IT 4.93 Which of the following fractions are equivalent to $\frac{-3}{5}$?

$\frac{-3}{-5}, \frac{3}{5}, -\frac{3}{5}, \frac{3}{-5}$

TRY IT 4.94 Which of the following fractions are equivalent to $-\frac{2}{7}$?

$\frac{-2}{-7}, \frac{-2}{7}, \frac{2}{7}, \frac{2}{-7}$

Fraction bars act as grouping symbols. The expressions above and below the fraction bar should be treated as if they were in parentheses. For example, $\frac{4+8}{5-3}$ means $(4+8) \div (5-3)$. The order of operations tells us to simplify the numerator and the denominator first—as if there were parentheses—before we divide.

We'll add fraction bars to our set of grouping symbols from Use the Language of Algebra to have a more complete set here.

Grouping Symbols

Parentheses	( )
Brackets	[ ]
Braces	{ }
Absolute value	| |
Fraction Bar	$\frac{\square}{\square}$

HOW TO

Simplify an expression with a fraction bar.

Step 1. Simplify the numerator.
Step 2. Simplify the denominator.
Step 3. Simplify the fraction.

EXAMPLE 4.48

Simplify: $\frac{4+8}{5-3}$.

Solution

	$\frac{4+8}{5-3}$
Simplify the expression in the numerator.	$\frac{12}{5-3}$

Simplify the expression in the denominator.	$\frac{12}{2}$
Simplify the fraction.	6

TRY IT 4.95 Simplify: $\frac{4+6}{11-2}$.

TRY IT 4.96 Simplify: $\frac{3+5}{18-2}$.

EXAMPLE 4.49

Simplify: $\frac{4-2(3)}{2^2+2}$.

Solution

	$\frac{4-2(3)}{2^2+2}$
Use the order of operations. Multiply in the numerator and use the exponent in the denominator.	$\frac{4-6}{4+2}$
Simplify the numerator and the denominator.	$\frac{-2}{6}$
Simplify the fraction.	$-\frac{1}{3}$

TRY IT 4.97 Simplify: $\frac{6-3(5)}{3^2+3}$.

TRY IT 4.98 Simplify: $\frac{4-4(6)}{3^3+3}$.

EXAMPLE 4.50

Simplify: $\frac{(8-4)^2}{8^2-4^2}$.

Solution

	$\frac{(8-4)^2}{8^2-4^2}$
Use the order of operations (parentheses first, then exponents).	$\frac{(4)^2}{64-16}$
Simplify the numerator and denominator.	$\frac{16}{48}$
Simplify the fraction.	$\frac{1}{3}$

TRY IT 4.99 Simplify: $\frac{(11-7)^2}{11^2-7^2}$.

TRY IT 4.100 Simplify: $\frac{(6+2)^2}{6^2+2^2}$.

EXAMPLE 4.51

Simplify: $\frac{4(-3)+6(-2)}{-3(2)-2}$.

Solution

	$\frac{4(-3)+6(-2)}{-3(2)-2}$
Multiply.	$\frac{-12+(-12)}{-6-2}$
Simplify.	$\frac{-24}{-8}$
Divide.	3

TRY IT 4.101 Simplify: $\frac{8(-2)+4(-3)}{-5(2)+3}$.

TRY IT 4.102 Simplify: $\frac{7(-1)+9(-3)}{-5(3)-2}$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Division Involving Mixed Numbers (https://www.openstax.org/l/24DivisionMixed)

Evaluate a Complex Fraction (https://www.openstax.org/l/24ComplexFrac)

SECTION 4.3 EXERCISES

Practice Makes Perfect

Multiply and Divide Mixed Numbers

In the following exercises, multiply and write the answer in simplified form.

176. $4\frac{3}{8} \cdot \frac{7}{10}$

177. $2\frac{4}{9} \cdot \frac{6}{7}$

178. $\frac{15}{22} \cdot 3\frac{3}{5}$

179. $\frac{25}{36} \cdot 6\frac{3}{10}$

180. $4\frac{2}{3}\left(-1\frac{1}{8}\right)$

181. $2\frac{2}{5}\left(-2\frac{2}{9}\right)$

182. $-4\frac{4}{9} \cdot 5\frac{13}{16}$

183. $-1\frac{7}{20} \cdot 2\frac{11}{12}$

In the following exercises, divide, and write your answer in simplified form.

184. $5\frac{1}{3} \div 4$

185. $13\frac{1}{2} \div 9$

186. $-12 \div 3\frac{3}{11}$

187. $-7 \div 5\frac{1}{4}$

188. $6\frac{3}{8} \div 2\frac{1}{8}$

189. $2\frac{1}{5} \div 1\frac{1}{10}$

190. $-9\frac{3}{5} \div \left(-1\frac{3}{5}\right)$

191. $-18\frac{3}{4} \div \left(-3\frac{3}{4}\right)$

Translate Phrases to Expressions with Fractions

In the following exercises, translate each English phrase into an algebraic expression.

192. the quotient of $5u$ and 11

193. the quotient of $7v$ and 13

194. the quotient of p and q

195. the quotient of a and b

196. the quotient of r and the sum of s and 10

197. the quotient of A and the difference of 3 and B

Simplify Complex Fractions

In the following exercises, simplify the complex fraction.

198. $\frac{\frac{2}{3}}{\frac{8}{9}}$

199. $\frac{\frac{4}{5}}{\frac{8}{15}}$

200. $\frac{-\frac{8}{21}}{\frac{12}{35}}$

201. $\frac{-\frac{9}{16}}{\frac{33}{40}}$

202. $\frac{-\frac{4}{5}}{2}$

203. $\frac{-\frac{9}{10}}{3}$

204. $\frac{\frac{2}{5}}{8}$

205. $\frac{\frac{5}{3}}{10}$

206. $\frac{\frac{m}{3}}{\frac{n}{2}}$

207. $\frac{\frac{r}{5}}{\frac{s}{3}}$

208. $\frac{-\frac{x}{6}}{-\frac{8}{9}}$

209. $\frac{-\frac{3}{8}}{-\frac{y}{12}}$

210. $\frac{2\frac{4}{5}}{\frac{1}{10}}$

211. $\frac{4\frac{2}{3}}{\frac{1}{6}}$

212. $\frac{\frac{7}{9}}{-2\frac{4}{5}}$

213. $\frac{\frac{3}{8}}{-6\frac{3}{4}}$

Simplify Expressions with a Fraction Bar

In the following exercises, identify the equivalent fractions.

214. Which of the following fractions are equivalent to $\frac{5}{-11}$?
$\frac{-5}{-11}, \frac{-5}{11}, \frac{5}{11}, -\frac{5}{11}$

215. Which of the following fractions are equivalent to $\frac{-4}{9}$?
$\frac{-4}{-9}, \frac{-4}{9}, \frac{4}{9}, -\frac{4}{9}$

216. Which of the following fractions are equivalent to $-\frac{11}{3}$?
$\frac{-11}{3}, \frac{11}{3}, \frac{-11}{-3}, \frac{11}{-3}$

217. Which of the following fractions are equivalent to $-\frac{13}{6}$?
$\frac{13}{6}, \frac{13}{-6}, \frac{-13}{-6}, \frac{-13}{6}$

In the following exercises, simplify.

218. $\frac{4+11}{8}$

219. $\frac{9+3}{7}$

220. $\frac{22+3}{10}$

221. $\frac{19-4}{6}$

222. $\frac{48}{24-15}$

223. $\frac{46}{4+4}$

224. $\frac{-6+6}{8+4}$

225. $\frac{-6+3}{17-8}$

226. $\frac{22-14}{19-13}$

227. $\frac{15+9}{18+12}$

228. $\frac{5\cdot 8}{-10}$

229. $\frac{3\cdot 4}{-24}$

230. $\frac{4\cdot 3}{6\cdot 6}$

231. $\frac{6\cdot 6}{9\cdot 2}$

232. $\frac{4^2-1}{25}$

233. $\frac{7^2+1}{60}$

234. $\frac{8\cdot 3+2\cdot 9}{14+3}$

235. $\frac{9\cdot 6-4\cdot 7}{22+3}$

236. $\frac{15\cdot 5-5^2}{2\cdot 10}$

237. $\frac{12\cdot 9-3^2}{3\cdot 18}$

238. $\frac{5\cdot 6-3\cdot 4}{4\cdot 5-2\cdot 3}$